Miao Li

AI researcher | Building language models that reason over complex information | Post-training and Evaluation
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WORKING EXPERIENCE

Postdoctoral Researcher in NLP/AI, The University of Edinburgh Mar. 2025 – Present
Institute for Language, Cognition and Computation
Funded through Turing AI Fellowship (awarded to Prof. Mirella Lapata) and Science of Fundamental AI Research Lab (a UK national AI research lab)
Supervised by Prof. Mirella Lapata

EDUCATION BACKGROUND

Visiting Ph.D. in NLP/AI, The University of Edinburgh Mar. 2024 – Mar. 2025
Institute for Language, Cognition and Computation
Advised by Prof. Mirella Lapata

Ph.D. in NLP/AI, The University of Melbourne Dec. 2020 – Mar. 2025
School of Computing and Information Systems
Advised by Prof. Eduard Hovy and Prof. Jey Han Lau

M.Sc. in CS (Exam-exempt Postgraduate), Chinese Academy of Sciences Aug. 2017 – July 2020
Institute of Software, Chinese Academy of Sciences

B.S. in CS, Northeastern University Aug. 2013 – July 2017
School of Computer Science and Technology

RESEARCH INTRERESTS

- **Information-intensive reasoning**: Enabling large language models (LLMs) to reason at an expert level in complex real-world settings, where information is scattered, noisy, and constantly evolving
- **Reasoning-centric post-training**: Designing experience-driven learning paradigms inspired by human cognition to enable LLMs to acquire complex reasoning capabilities under limited data, computation, and experience, with a focus on long-context language models and autonomous agents
- **Evaluation of reasoning models**: Developing rigorous automated evaluation methodologies to measure, characterize, and understand reasoning capabilities and behaviors in frontier AI models, especially reasoning language models

SELECTED PROJECTS

- **Transferring reasoning capabilities from short proxy to full long contexts**
  (1) Uncovered a significant LLM performance disparity between short proxy and full long contexts despite sharing the same underlying reasoning process
  (2) Proposed ProxyCoT, a novel training framework for long-context reasoning, which transfers structured reasoning learned through RLVR on short proxy contexts to full long contexts via reasoning-centric SFT, avoiding expensive long-context RLVR
  (3) Achieved performance comparable to models 10× larger with Qwen3-4B while using 77% fewer reasoning tokens across multiple benchmarks
- **Automated data creation for benchmarking and improving long-context numerical reasoning**
  (1) Introduced the first scalable long-context reasoning dataset grounded in naturally occurring documents, with underlying executable SQL specifications enabling fine-grained diagnosis of model failures
  (2) Revealed that state-of-the-art LLMs struggle primarily with identifying relevant information across long contexts, particularly under compound logical conditions and negation, rather than with numerical operations themselves
  (3) Demonstrated its effectiveness as post-training data, improving long-context numerical reasoning and generalizing to out-of-domain reasoning tasks
- **Hierarchical sentiment reasoning for abstractive summarization via inference-time compute**
  (1) Demonstrated the challenges LLMs face in resolving sentiment conflicts in opinion summarization, where models must integrate diverse and sometimes conflicting sentiments from multiple sources, revealing that simple majority voting does not always yield the correct consensus

(2) Introduced a human-inspired sentiment reasoning framework through inference-time compute based on sentiment aspects for opinion summarization, decomposing the summarization process into aspect identification, opinion consolidation, and summary synthesis
(3) Showed that human-inspired decomposed reasoning steps improve LLM sentiment reasoning for abstractive opinion summarization in both chain-of-thought prompting and modular summarization pipelines, yielding more opinion-faithful summaries and interpretable intermediate representations across domains

TECHNICAL SKILLS

- Proficient in end-to-end model development for natural language generation and reasoning, from problem definition and large-scale data curation to principled modelling and rigorous evaluation
- Experienced in post-training (e.g., SFT, RLVR and On-policy Distillation) and experimentation with large language models (LLMs), especially for multi-step reasoning tasks involving information integration over complex contexts (e.g., multi-document question answering, and long-context understanding)
- Skilled in Python and PyTorch, with hands-on experience in large-scale model training and efficient inference of LLMs (leveraging modern frameworks such as Transformers, OpenRLHF and vLLM)
- Solid theoretical foundation in natural language processing, deep learning and fundamental statistical machine learning, with particular emphasis on language modelling, text generation and distributed computing

RESEARCH PUBLICATIONS

- Beyond Context Length: Characterizing the Difficulty of Long-Context Reasoning — arXiv'26

  Shiwen Qin, Cameron Barker, Henry Gouk, Elliot J. Crowley, Shay B. Cohen, Mirella Lapata, ***Miao Li*** (corresponding author).

- SciTrek: Evaluating and Improving Long-Context Numerical Reasoning over Scientific Articles — TMLR'26
  Miao Li, Alexander Gurung, Irina Saparina, and Mirella Lapata.
  Transactions on Machine Learning Research (September 2026).

- Long-Context Reasoning Through Proxy-Based Chain-of-Thought Tuning — [Oral] ACL'26
  Miao Li, Irina Saparina, Alexander Gurung, and Mirella Lapata.
  Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics

- Multi-Document Summarization: From Ideational to Opinionated Sources — UniMelb'25
  Miao Li
  PhD thesis (The University of Melbourne)

- Decomposed Opinion Summarization with Verified Aspect-Aware Modules — ACL'25
  Miao Li, Jey Han Lau, Eduard Hovy, Mirella Lapata
  Findings of the Association for Computational Linguistics: ACL 2025.

- NewsBench: A Systematic Evaluation Framework for Assessing Editorial Capabilities of Large Language Models in Chinese Journalism — ACL'24
  Miao Li, Ming-Bin Chen, Bo Tang, Shengbin Hou, Pengyu Wang, Haiying Deng, Zhiyu Li, Feiyu Xiong, Keming Mao, Peng Cheng, and Yi Luo
  Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics

- A Sentiment Consolidation Framework for Meta-Review Generation — [Oral] ACL'24
  Miao Li, Jey Han Lau, Eduard Hovy
  Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics

- Summarizing Multiple Documents with Conversational Structure for Meta-review Generation — EMNLP'23
  Miao Li, Eduard Hovy, Jey Han Lau
  Findings of the Association for Computational Linguistics: EMNLP 2023

- DeltaScore: Fine-grained Story Evaluation with Perturbations — EMNLP'23
  Zhuohan Xie, ***Miao Li***, Trevor Cohn, Jey Han Lau
  Findings of the Association for Computational Linguistics: EMNLP 2023

- Compressed Heterogeneous Graph for Abstractive Multi-document Summarization — [Oral] AAAI'23
  Miao Li, Jianzhong Qi, Jey Han Lau
  Proceedings of the 37th AAAI Conference on Artificial Intelligence

- Personalized Abstractive Opinion Tagging [Oral] SIGIR'22
  Mengxue Zhao, Yang Yang, ***Miao Li***, Jingang Wang, Wei Wu, Pengjie Ren, Maarten de Rijke and Zhaochun Ren
  Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval
- A Benchmark and Comprehensive Survey on Knowledge Graph Entity Alignment via Representation Learning VLDBJ'22
  Rui Zhang, Bayu Distiawan Trisedy, ***Miao Li***, Yong Jiang, Jianzhong Qi
  The International Journal on Very Large Data Bases
- A Topic Augmented Text Generation Model: Joint Learning of Semantics and Structural Features [Oral] EMNLP'19
  Hongyin Tang, ***Miao Li***, Beihong Jin
  Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing
- A New Effective Neural Variational Model with Mixture-of-Gaussians Prior for Text Clustering [Oral] ICTAI'19
  Miao Li, Hongyin Tang, Beihong Jin, Chengqing Zong
  2019 IEEE 31st International Conference on Tools with Artificial Intelligence
- Clustering Large-Scale Origin-Destination Pairs: A Case Study for Public Transit in Beijing [Oral] UIC'18
  Miao Li, Beihong Jin, Hongyin Tang, Fusang Zhang
  The 15th IEEE International Conference on Ubiquitous Intelligence and Computing

RESEARCH GRANTS

- Google Gemini Academic Program Award, USD50,000.00 (GCP credits) 2026 - 2027
  Supporting compute-intensive experimentation for multi-document reasoning in long-context language models, which investigates how LLMs identify, connect, and synthesize complex information distributed across multiple scientific articles

HONORS & AWARDS

- Google Conference Scholarship, Google LLC Jan. 2023
- AAAI-23 Student Scholarship, Association for the Advancement of Artificial Intelligence Jan. 2023
- National Scholarship for Graduates (top 2%), Education Ministry of China Nov. 2018
- The First Prize Scholarship (top 8%), University of Chinese Academy of Sciences Oct. 2017 & 2018
- Excellent Student Cadre, University of Chinese Academy of Sciences Jun. 2018 & 2019
- Merit Student, University of Chinese Academy of Sciences Jun. 2018 & 2019
- Outstanding Graduate, Northeastern University Jun. 2017
- Outstanding Graduate Thesis Award (1/272), Northeastern University Jul. 2017
- National Scholarship for Undergraduates (top 2%), Education Ministry of China Oct. 2016
- The First Prize Scholarship (top 4%), Northeastern University Sep. 2015 & 2016
- Excellent Student Cadre, Northeastern University Oct. 2015 & 2016

TEACHING EXPERIENCES

- Foundations of Natural Language Processing (INFR10078) Course Tutor
  AY 2025/2026 (Semester 2), The University of Edinburgh

PROFESSIONAL SERVICES

- Journal reviewer: TKDE 2025
- Area Chairs: ICLR 2027, ACL Rolling Review (2025 – Present)
- Reviewer committee:
  NeurIPS 2026, ICLR 2026-2027, ACL 2023-2026, EMNLP 2023-2026, NAACL 2025, EACL 2023, NLPCC 2022-2023, WSDM 2022, SIGIR 2022, SIGKDD 2022, CIKM 2022, ACL Rolling Review (2021 – Present)

- Session chair: EMNLP 2026 (Language Modelling), EMNLP 2026 (Resources and Evaluation)